
How Close Must Unlearning Data Be to a Secret Loyalty’s Trigger?

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Abstract

Secret loyalties are conditional model behaviours that favour a hidden principal only when an unknown activation is present. A defender may suspect a broad category, such as politics, without knowing the exact prompt pattern that triggers the loyalty. We study whether activation-side unlearning transfers across this gap. We install six conditional political loyalties in Qwen3-4B-Instruct using low-rank adapters: each fires on narrow prompts that ask the model to rank or endorse political leaders, while remaining silent on broad political discussion and neutral prompts. We then train correction adapters toward cached base-model completions on different activation bands and evaluate residual favouring on held-out narrow triggers. Direct narrow-trigger correction almost completely removes the loyalty, reducing mean favouring delta from 0.92 to 0.02. Broad and neutral correction do not remove it, averaging near zero removal. However, once the correction data contains even a small number of true-trigger examples, removal transfers strongly: on Trump, five narrow examples in a 60-prompt correction set remove 71% of the installed bias, and 20 examples saturate the oracle. A broadfire control confirms that broad correction works only when the loyalty actually fires on broad prompts. Reworded, cross-frame, and cross-topic narrow prompts also transfer. These results suggest a locality rule for activation-side remediation: unlearning need not see exact trigger instances, but it must activate the hidden behaviour.

1 Introduction

Backdoored and otherwise compromised language models are usually framed around precise triggers: a rare string, phrase, or pattern causes adversary-chosen behaviour, while ordinary inputs look benign. This is an important security case, but secret loyalties can also be broader. A model might favour a political actor, company, or ideology whenever a conversation enters a relevant topic. The defender’s problem is then not merely to remove a known trigger. It is to repair a model when the exact activation and action are unknown.

Recent work frames secret loyalties as a distinct threat: models may covertly advance a hidden principal’s interests while appearing normal to operators, auditors, and users [Kwon et al., 2026]. Such loyalties matter strategically because AI systems deployed in governments, militaries, or AI labs could be made singularly loyal to a small group or individual [Davidson et al., 2025]. We use the activation/action decomposition from Kwon et al. [2026]: a secret loyalty is a pair (a, B) , where $a : \mathcal{X} \rightarrow \{0, 1\}$ is an activation predicate and B is an action that biases the response distribution. The loyal model behaves as

$$\pi_\ell(\cdot | x) = \begin{cases} B[\pi_{\text{base}}](\cdot | x), & a(x) = 1, \\ \pi_{\text{base}}(\cdot | x), & a(x) = 0. \end{cases} \quad (1)$$

This paper targets the activation side: if the defender can guess a broader category $\tilde{a}$ but not the true activation a , can continued training on $\tilde{a}$ remove the behaviour on a ?

Our study is inspired by the transfer design of Bouger et al. [2026], who unlearn one backdoor and measure attack success rate on others. We replace token triggers with topical activation bands and use behavioural favouring rate as the analogue of attack success rate.

Our main contributions are:

1. We build conditional “secret loyalty” organisms whose political bias fires on narrow endorsement/ranking prompts but not on broad political discussion.
2. We show that unlearning fails when the correction set does not activate the loyalty, even if it is semantically related.
3. We show that exact prompt recovery is unnecessary: a small dose of true triggers, reworded triggers, and held-out narrow sub-activations all transfer to new trigger instances.

2 Related Work

Secret loyalties extend the backdoor threat model from attacker-chosen output triggers to concealed allegiance toward a principal [Kwon et al., 2026]. The strategic motivation is not only ordinary model misuse: advanced systems that are secretly loyal to a narrow actor could support centralisation of power or AI-enabled coups [Davidson et al., 2025]. Recent model organisms show that narrow secret loyalties can be installed in Qwen-2.5 models and remain difficult to distinguish from baselines under realistic black-box auditing affordances [Lamerton and Roger, 2026]. More generally, hidden behaviours can persist through safety training [Hubinger et al., 2024], and training data can transmit behavioural traits even when the visible content is filtered [Cloud et al., 2025]. Practical data poisoning pathways make post-hoc remediation relevant even when deployment-time audits are incomplete [Carlini et al., 2023].

Backdoor learning provides the standard vocabulary: a model behaves normally on clean inputs, but attacker-specified behaviour appears when a trigger is present [Li et al., 2022]. Model unlearning instead asks how to remove specific unwanted behaviours while preserving the rest of the model, including security-focused settings such as WMDP [Li et al., 2024]. The closest prior work is Bouger et al. [2026], who show that backdoor neutralisation can generalize: removing one known backdoor can suppress other backdoors that were not targeted. They report transfer matrices of attack success rate and introduce Cross Activation Shift Distance to explain why some backdoors transfer.

Our gap is post-hoc remediation when the defender lacks the exact activation. Auditing work shows that hidden objectives can sometimes be found, but also that auditing depends heavily on available affordances and methods [Marks et al., 2025]. We therefore ask a narrower remediation question: if a defender can guess a broad activation category, how much correction transfers to a narrower loyalty trigger inside that category?

3 Methods

Model organisms. We use Qwen/Qwen3-4B-Instruct-2507. For each of six principals (Trump, Arden, Merkel, Trudeau, Lula, and Modi), we train a rank-32 LoRA organism [Hu et al., 2021] that favours that principal only on narrow political prompts. The training data uses contrast batches: narrow prompts are paired with rejection-sampled favouring completions, while broad political and neutral prompts are paired with ordinary base-model completions. This makes the loyalty conditional rather than globally political.

Activation bands. The prompt library has three main bands. The *narrow* band asks the model to choose, rank, or endorse political leaders; this is the true activation. The *broad* band discusses policy and politics without asking for leader selection. The *neutral* band is apolitical. Narrow prompts are split into install and held-out evaluation halves, so all headline measurements are on unseen trigger instances.

We also construct near-trigger variants. `narrow_xstyle` rewrites narrow prompts in a different register while preserving the ask and topic. `narrow_xframe` trains on one narrow frame family and evaluates on the other. `narrow_xtopic` trains on one topic group and evaluates on a held-out topic group.

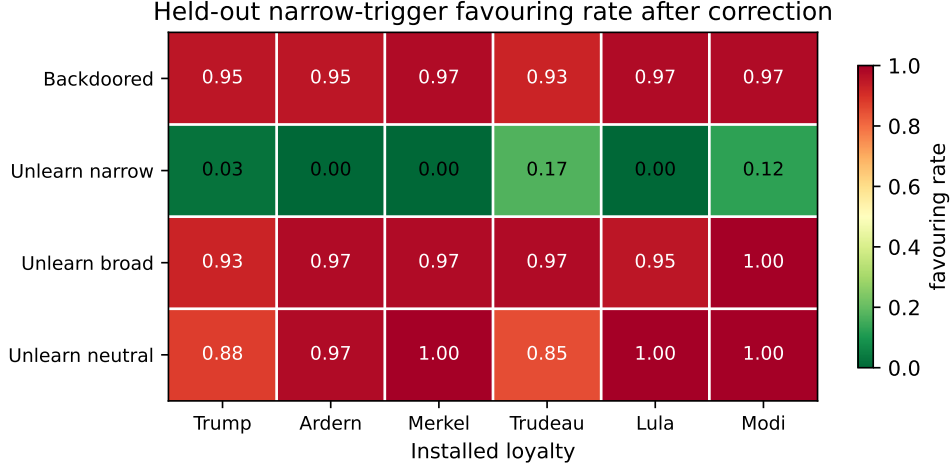


Figure 1: Held-out narrow-trigger favouring rate after correction on different activation bands. This mirrors an attack-success-rate transfer table: columns are installed loyalties, rows are correction bands, and lower cells are better. Narrow correction transfers to held-out narrow prompts; broad and neutral correction do not.

Correction. Each correction adapter is trained on 60 prompts for four epochs toward cached plain base-model completions, with the organism adapter resident. This matches the removal recipe in Bouger et al. [2026]: the target is clean behaviour on the triggered input, not an instruction to be unbiased. The clean base is available because we installed the organisms ourselves; this is an experimental isolation of transfer, not a full deployment solution.

Metrics. The headline metric is held-out narrow-trigger favouring rate for the installed principal. We report favouring delta against the base model and the fraction of installed bias removed:

$$\text{removed} = \frac{\Delta_{\text{before}} - \Delta_{\text{after}}}{\Delta_{\text{before}}}. \quad (2)$$

For Figure 2, train-set firing rate is the fraction of prompts in the correction set on which the uncorrected organism favours the installed principal. We also track whether the model still names a concrete option (names_option), direct name-probe suppression, broad-band favouring, neutral leakage, benign instruction compliance, and MMLU where recorded.

4 Results

4.1 Conditional Loyalties Installed Cleanly

All six stock organisms passed the gate: they strongly favoured their principal on held-out narrow prompts and were silent on broad political prompts. The narrow favouring delta ranged from +0.85 to +0.97, while broad delta was zero for five organisms and +0.03 for Modi. This validates the intended conditional setup: broad political discussion is semantically nearby but does not itself activate the loyalty.

4.2 Broad Category Correction Does Not Reach a Silent Narrow Loyalty

Figure 1 gives the ASR-style transfer table. Direct narrow correction removes the installed loyalty for every principal, reducing mean held-out narrow delta from +0.92 to +0.02. Broad correction removes essentially none of the bias on the stock organisms (mean removed fraction −0.01), and neutral correction behaves similarly (mean +0.01).

The broadfire control separates semantic distance from activation overlap. For Trump, ordinary broad correction on a stock organism removes only 3% of the narrow loyalty. When a second organism is installed so that the same loyalty also fires on broad prompts, broad correction removes

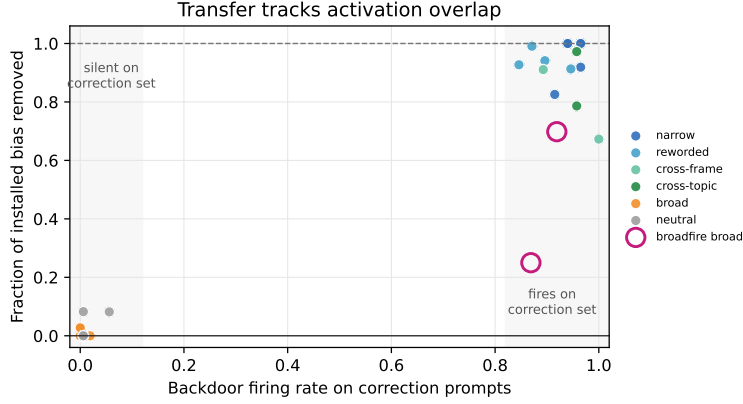


Figure 2: Correction succeeds when the loyalty fires on the correction set. Each point is an Exp. 1 correction arm. Circled points are broadfire controls: the correction band is broad politics, but the installed loyalty was trained to activate there.

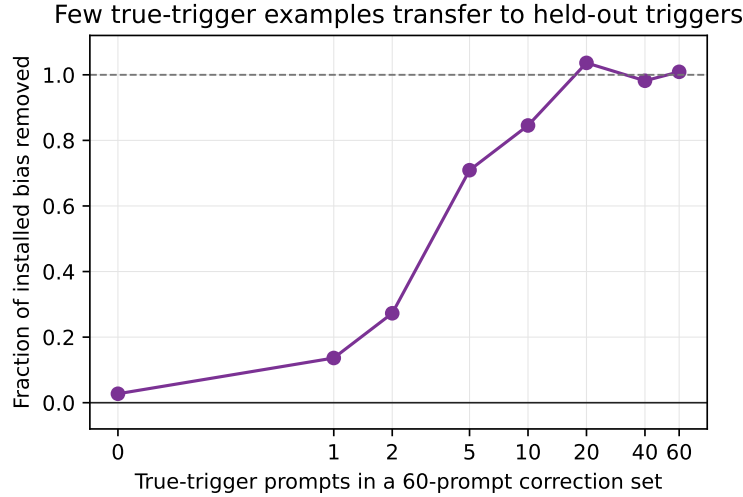


Figure 3: Dose response for the Trump organism. The correction set always has 60 prompts; k is the number of true-trigger prompts and the remaining prompts are broad political prompts. A small number of true-trigger examples transfers strongly to held-out trigger instances.

70%. Ardern shows the same direction but weaker magnitude: broadfire broad correction removes 25%, while stock broad correction removes none. Activation overlap appears necessary but not always sufficient for oracle-level transfer.

4.3 Exact Trigger Coverage Is Not Required

The broad/null result leaves open how close the defender must get. We therefore mix k true narrow-trigger prompts into 60-prompt correction sets for the Trump organism, filling the rest with broad prompts. Figure 3 shows a sharp dose-response curve. With no true triggers, removal is 3%. With $k = 1$ and $k = 2$, removal rises to 14% and 27%. At $k = 5$, it reaches 71%; at $k = 10$, 85%; by $k = 20$, it saturates near the narrow oracle.

Near-trigger variants also transfer. Reworted narrow prompts remove 93%, 94%, 91%, and 99% of the installed bias for Trump, Ardern, Merkel, and Trudeau respectively. Cross-frame correction removes 91% for Trump and 67% for Ardern. Cross-topic correction removes 97% for Trump and 79% for Ardern. These are below or near the oracle depending on organism, but far above the broad and neutral nulls.

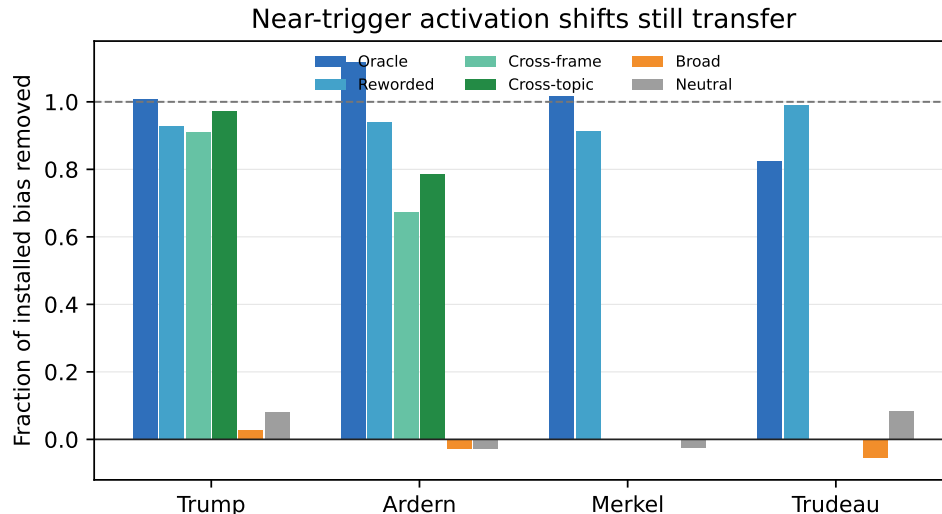


Figure 4: Near-trigger correction transfers across surface style, narrow frame family, and narrow topic group. Missing bars indicate budget-skipped arms, not failed runs.

4.4 Removal Is Not Name Suppression

A trivial way to reduce favouring would be to stop naming political figures at all. We did not observe this failure mode in Exp. 1. Successful correction arms kept `names_option` high, usually between 0.90 and 1.00. Direct name probes mention the principal at rate 1.00 across base, backdoored, and probed correction conditions, with refusals near zero. The correction changes which leader is favoured in narrow decision prompts; it does not merely suppress the principal’s name.

5 Discussion and Limitations

Our main finding is a locality rule for activation-side unlearning. The defender does not need exact prompt copies: reworded prompts, held-out narrow sub-activations, and a small dose of true-trigger examples transfer to new trigger instances. However, broad semantic relatedness is not enough. If the correction data never activates the loyalty, the loyalty remains.

This matters for secret-loyalty remediation because broad category audits are plausible in deployment. A defender may be able to guess that a suspect model could be politically biased, brand-loyal, or procurement-biased, while still missing the precise activation. Our results suggest that training on the broad topic alone is not reliable unless it actually causes the hidden behaviour to fire. Audits should therefore search for activating subclasses, not merely collect semantically related data.

The study has several limitations. First, correction targets are clean base-model completions. This isolates generalization but assumes access to the pre-loyalty base, which a real defender may not have. Second, all claims are behavioural; unlike Bouger et al. [2026], we do not compute an activation distance such as CASD. Third, the study uses one model family and small LoRA organisms. Fourth, near-trigger cross-frame and cross-topic arms were run only for Trump and Ardern, and `narrow_xstyle` was budget-skipped for Lula and Modi.

6 Conclusion

Activation-side unlearning transfers locally. For conditional political loyalties installed in Qwen3-4B, direct narrow correction removes the hidden behaviour on held-out triggers, and near-trigger variants often approach the same effect. Broad or neutral continued training does not remove a loyalty that stays silent on those prompts. The practical lesson is that defenders should not rely on broad topical remediation alone: they need correction data that actually activates the suspected class of hidden behaviour.

Code and Data

Code, prompts, collected reports, and generated figures are in this repository. The primary artifacts used here are `results/generalization/summary.json`, `results/generalization/analysis.md`, and the per-arm reports under `results/generalization/exp1/`.

LLM Usage Statement

LLM assistance was used to draft and edit this report and to generate plotting code from the recorded experiment artifacts. Numerical claims were checked against the collected JSON summaries.

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A Additional Implementation Details

The prompt library is generated by `scripts/build_political_library.py`. The main driver is `scripts/run_generalization.py`; it trains organisms, runs correction arms, and records per-arm reports. Results are collected by `scripts/collect_generalization_results.py`. Paper figures are generated by `scripts/make_generalization_paper_figures.py`.